

Unit 5.4 The Fundamental Theorem of Calculus p. 396-399

Note Title

2016/05/17

p396

The Fundamental Theorem of Calculus, Part 2 If f is continuous on $[a, b]$, then

$$\int_a^b f(x) dx = F(b) - F(a)$$

where F is any antiderivative of f , that is, a function such that $F' = f$.

Proof: Divide $[a, b]$ into n subintervals of equal length $\Delta x = \frac{b-a}{n}$.

Let $x_0 = a < x_1 < \dots < x_{n-1} < x_n = b$ be the endpoints of the interval. (A)

Let F be an antiderivative of f so that $F' = f$ (B)

Therefore:

- F is differentiable on $[a, b]$ and
- F is continuous on $[a, b]$

$[f \text{ diff} \Rightarrow f \text{ continuous}] [a, b]$

Use the MVT on each subinterval $[x_{i-1}, x_i]$

1. F is continuous on $[x_{i-1}, x_i]$

2. F is differentiable on (x_{i-1}, x_i)

therefore by the MVT there exists a number $c_i \in (x_{i-1}, x_i)$ such that

$$\frac{F(x_i) - F(x_{i-1})}{x_i - x_{i-1}} = F'(c_i) = f(c_i) \quad (C) \quad \text{from (B).}$$

Let these c_i 's be the sample point in each interval.

As f is continuous on $[a, b]$, f is integrable on $[a, b]$ - Theorem p 373

Using definition p 378.

$$\begin{aligned} \int_a^b f(x) dx &= \lim_{n \rightarrow \infty} \sum_{i=1}^n f(c_i) \Delta x \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{F(x_i) - F(x_{i-1})}{x_i - x_{i-1}} \Delta x \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{F(x_i) - F(x_{i-1})}{x_i - x_{i-1}} (x_i - x_{i-1}) \quad \text{From C} \\ &= \lim_{n \rightarrow \infty} \left(\sum_{i=1}^n [F(x_i) - F(x_{i-1})] \right) \\ &= \lim_{n \rightarrow \infty} \left[\cancel{F(x_1)} - F(x_0) + F(x_2) - \cancel{F(x_1)} + \dots \right. \\ &\quad \left. + F(x_n) - F(x_{n-1}) \right] \\ &= \lim_{n \rightarrow \infty} [F(x_n) - F(x_0)] \\ &= \lim_{n \rightarrow \infty} [F(b) - F(a)] \\ &= F(b) - F(a) \end{aligned}$$

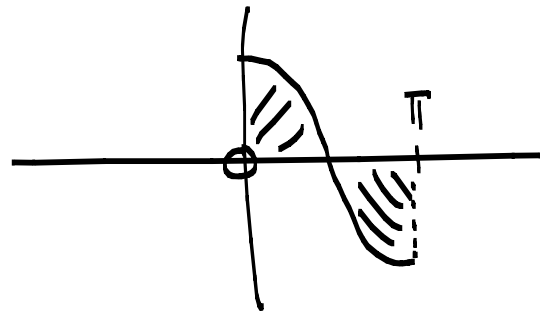
Examples:

1. $\int_0^2 \sqrt{4-x^2} dx$? Geometry.

2. $\int_0^\pi \cos 3x dx$

$$= \frac{1}{3} \sin 3x \Big|_0^\pi$$

$$= \frac{1}{3} \sin 3(\pi) - \frac{1}{3} \sin 3(0) = 0$$



2. $\int_{-1}^2 e^{5x} dx$ [definite integral].

$$= \frac{1}{5} e^{5x} \Big|_{-1}^2 = \frac{1}{5} e^{10} - \frac{1}{5} e^{-5}$$

3. $\int_{-2}^1 x^2 e^{x^3} dx$

4. $\int_0^1 2^t \cos 2^t dt$

5. $\int_{-2}^{-1} \sqrt{1-x} dx$

6. $\int_{-1}^1 x \sqrt{1+x^2} dx$